

Mentored Research Grant (MRG)

Project Title

Experiment-Driven Design of Multi-Material 3D-Printed Tensegrity Structures for Energy Absorption

Principal Investigator(s)

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PI Department: Mechanical Engineering

Co-PI: Sterling G. Baird, Department of Mechanical Engineering

Abstract

We propose a two-year experiment-driven mentored research program in which 2 undergraduate students design, fabricate, and test multi-material 3D-printed tensegrity-inspired structures for energy absorption. Using PLA for rigid struts and TPU for flexible tension elements on a single FDM platform, students will rapidly print and test many candidate geometries, generating high-quality physical data. A Bayesian optimization loop operating directly on experimental measurements will guide the search toward optimal energy-absorbing designs. The project is structured around layered mentoring—weekly one-on-one faculty meetings, lab group sessions, and graduate co-mentoring—that progressively moves students from guided tasks to independent research ownership. Students will gain hands-on experience in CAD design, multi-material 3D printing, mechanical testing, and data-driven optimization. Expected outcomes include undergraduate conference presentations (UCUR, ASME IDETC), a peer-reviewed journal submission, and development of transferable technical skills. This project provides a rich mentored learning environment at the intersection of structural mechanics, advanced manufacturing, and machine learning.

Number of Students to be Mentored

Number of Undergraduate Students: 2

Number of Graduate Students: 1 (co-mentor, in support of creating the mentoring environment)

Budget

Item	Year 1	Year 2	Total
Undergraduate wages	\$9,000	\$9,000	\$18,000
Graduate wages	\$1,250	\$1,250	\$2,500
Supplies	\$2,250	\$750	\$3,000
Travel	\$0	\$1,500	\$1,500
Total	\$12,500	\$12,500	\$25,000

Relationship to External Funding

The PIs do not currently hold external funding for tensegrity or architected materials research. This MRG provides the primary support for establishing a mentored undergraduate research program in this area. If successful, the preliminary data may inform a future NSF proposal; however, the MRG is self-contained and its outcomes do not depend on external funding.

1 Research Motivation and Overview

We propose a two-year **experiment-driven** mentored research program in which 2 undergraduate students design, fabricate, and test **multi-material 3D-printed tensegrity-inspired structures** for energy absorption. Tensegrity (tensional integrity) structures—assemblies of rigid compression members suspended within a continuous network of tension members—exhibit remarkable strength-to-weight ratios and tunable mechanical responses, making them attractive for protective equipment, packaging, and aerospace applications [Skelton and de Oliveira, 2009, Fraternali et al., 2015]. Recent work has shown that *tensegrity-inspired* architectures can be directly 3D-printed and retain desirable load-limiting behavior under impact [Pajunen et al., 2019]. Because multimaterial FDM TPU elements are not ideal cables, we target tensegrity-inspired architectures that reproduce key tensegrity-like mechanical responses while remaining manufacturable; notably, TPU’s rate-dependent damping is an *asset* for energy absorption rather than merely a deviation from idealized behavior.

Multi-material 3D printing with PLA (rigid struts) and TPU (flexible tension elements) enables rapid fabrication of diverse tensegrity-inspired geometries on a single print platform [Ye et al., 2023, Khatri and Egan, 2024]. Rather than relying on costly simulation campaigns, we leverage this manufacturing agility to pursue a **high-throughput experimental approach**: students will print, test, and iterate on many candidate designs—we estimate 50–100+ unique specimens over two years—generating high-quality physical data directly. **Bayesian optimization** (BO) will guide the experimental search, selecting the most informative designs to print and test next based on prior results [Shahriari et al., 2016, Frazier, 2018]. This experiment-first strategy produces reliable, real-world performance data while keeping the project accessible to undergraduates at all levels—students can contribute meaningfully through hands-on printing and testing without requiring advanced modeling expertise.

A primary objective of this project is to provide a rich, interdisciplinary mentored research experience for undergraduates. Through weekly one-on-one mentoring, structured lab meetings, and a progressive scaffolding model that moves students from guided tasks to independent research ownership, each student will gain hands-on skills in CAD, multi-material 3D printing, mechanical testing, and data-driven optimization. Each student will be responsible for defining design variables, executing experimental or computational campaigns, analyzing data, presenting results, and drafting manuscript sections.

2 Background

Tensegrity structures consist of isolated compression members (struts) held in equilibrium by a continuous network of tension members [Skelton and de Oliveira, 2009, Sultan, 2009]. Their defining characteristic—no two rigid bars touch—yields lightweight assemblies with surprising load-bearing capacity and tunable nonlinear responses [Amendola et al., 2014, Zhang et al., 2015]. Pajunen et al. [Pajunen et al., 2019] demonstrated that *tensegrity-inspired* unit cells can be directly 3D-printed and exhibit load-limiting plateaus under drop-weight impact. Multi-material rigid-soft printing [Ye et al., 2023, Khatri and Egan, 2024] has shown that PLA–TPU combinations printed on a single FDM platform achieve tunable energy absorption. Ye et al. demonstrated a wrapping-based strategy (rigid cores encapsulated by continuous soft skins) that prevents interface delamination and enables cyclic durability—a critical enabler for our fabrication approach. These

manufacturing advances mean that undergraduates can design, print, and test many configurations in rapid succession, making 3D printing the natural engine of experimental exploration.

Bayesian optimization (BO) is a sequential, model-based strategy for optimizing expensive black-box functions using a Gaussian process surrogate [Shahriari et al., 2016, Frazier, 2018]. BO has driven breakthroughs across domains—from tuning hyperparameters of AlphaGo [Silver et al., 2016] to optimizing sustainable concrete formulations [Ament et al., 2023] and discovering state-of-the-art organic laser emitters [Strieth-Kalthoff et al., 2024]—and has been applied to metamaterial design [Wang et al., 2022]. In our framework, BO operates *directly on experimental data*: each round of printing and testing provides observations that update a Gaussian process surrogate, which then recommends the next set of designs to fabricate. This closed-loop print–test–optimize cycle eliminates the need for simulation as a prerequisite and keeps the workflow centered on physical experimentation—an approach that is both scientifically rigorous and naturally suited to mentoring undergraduates at all experience levels.

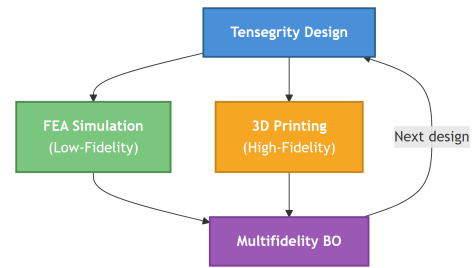


Figure 1: Research framework overview.

3 Student Research Project 1: Design, Fabrication, and Testing

Student profile: One sophomore or junior undergraduate. No prior simulation or advanced modeling experience is required; the hands-on nature of this project makes it an ideal entry point for students early in their studies.

Scope and Deliverables

This student will lead the design-and-build cycle that forms the experimental backbone of the project. Working under weekly one-on-one mentoring from Prof. Hill, the student will begin by **parameterizing a family of tensegrity-inspired unit cells** in CAD (SolidWorks or Fusion 360), defining key variables—strut diameter and length (PLA), tension-element cross-section (TPU), connectivity topology, and unit-cell tiling—that span the design space. Following Ye et al. [2023], the designs use a core-wrapping strategy (rigid PLA struts encapsulated by continuous TPU skins) to ensure robust mechanical interlocking and prevent interface delamination. Because PLA–TPU interfaces are known to exhibit mixed-mode fracture behavior and potential void formation, print parameter validation and interface robustness will be explicitly evaluated during early testing.

The student will then **fabricate structures** using the department’s multi-material FDM printer, systematically iterating on print parameters (layer height, infill, temperature) to achieve reliable geometries in both PLA and TPU. By printing many candidate designs quickly, the student builds an expanding library of tested specimens and develops practical expertise in additive manufacturing—a transferable skill valued in both graduate research and industry. The student works closely with Student 2, who tests each batch and returns performance data; this tight collaboration teaches both students the value of interdisciplinary teamwork and creates natural peer-mentoring opportunities as each develops complementary expertise. Throughout, Prof. Baird’s master’s student serves as a day-to-day co-mentor, guiding the student through equipment operation, troubleshooting print fail-

ures, and interpreting early results, so that learning happens at multiple cadences: weekly faculty meetings for strategy and daily peer support for execution.

Mentoring outcomes: The student will develop skills in CAD, multi-material additive manufacturing, and systematic experimental design. Structured mentoring scaffolds move the student from guided replication of existing designs to independent creation of novel geometries by mid-Year 1. The student will present results at **UCUR** or **ASME IDETC** and contribute as co-author on a journal manuscript.

4 Student Research Project 2: Mechanical Testing and Data-Driven Optimization

Student profile: One sophomore, junior, or senior undergraduate with coursework or interest in mechanics of materials or data analysis.

Scope and Deliverables

This student will lead physical testing and the Bayesian optimization campaign, working under weekly one-on-one mentoring from Prof. Baird. The work begins with **conducting experimental testing** on structures fabricated by Student 1: quasi-static compaction tests and drop-weight impact tests following the protocol of Pajunen et al. [2019]. Primary metrics include *peak transmitted force*, *specific energy absorption* (SEA), and *compaction efficiency*—tangible, operationally measurable outcomes well suited to undergraduate research.

With an initial data set in hand, the student will **implement a single-fidelity Bayesian optimization loop** (BoTorch/Ax) that operates directly on experimental measurements. After each BO iteration the surrogate recommends the next batch of designs; the student coordinates with Student 1 to print and test them, closing the loop. This closed-loop *design–print–test–optimize* cycle teaches the student data-driven decision-making in a concrete, physical context. If time and student interest permit, lightweight simulations may be added as an optional low-fidelity data source to accelerate convergence [Mo et al., 2023], but the campaign does not depend on simulation fidelity.

Prof. Baird’s master’s student provides day-to-day co-mentoring in Python, data analysis, and BO implementation, creating a layered mentoring structure. In Year 2, both students mentor newly recruited undergraduates, reinforcing their own understanding while sustaining the research pipeline.

Mentoring outcomes: The student will develop skills in experimental mechanics (test design, data acquisition), scientific computing (Python), and data-driven optimization. They will present results at **ASME IDETC** or **UCUR** and contribute as co-author on a journal manuscript.

5 Mentoring Environment

Each student in this program follows a structured path from guided exploration to independent research ownership, supported by layered mentoring at every stage. Because the project is centered on hands-on experimentation—designing, printing, and testing physical structures—students

engage with tangible problems from day one, building confidence and competence through rapid, concrete feedback loops.

Recruitment

Students will be recruited from courses in mechanics of materials, machine design, kinematics, and compliant mechanisms, targeting sophomores through seniors. The experiment-driven nature of the project lowers the barrier to entry: students need not have prior simulation or programming experience, broadening the pool of potential mentees.

Mentoring Structure

Students meet individually with Profs. Hill and Baird each week (30 min) for technical guidance and professional development, and participate in weekly lab group meetings (1 hr) that include progress updates, student literature presentations, and skill-building workshops covering scientific writing, presentation skills, and research ethics. Prof. Baird’s master’s student serves as a graduate co-mentor, providing day-to-day guidance in 3D printing, testing procedures, coding, and data interpretation; this layered model ensures timely support while the graduate student develops their own mentoring skills. Students begin with structured tasks—replicating known tensegrity geometries and running standard test protocols—and progress to independent design decisions, culminating in ownership of their research project. Year 1 focuses on technical skill acquisition and guided execution; Year 2 emphasizes independent hypothesis generation, experimental design, and peer mentoring. In Year 2, experienced students mentor newly recruited sophomores, reinforcing their own knowledge and creating a sustainable pipeline of trained researchers.

Student Development Outcomes

The experiment-driven workflow structures mentoring around concrete milestones: a student’s first successful multi-material print, first compression test, first data set fed into the BO loop. On the *technical* side, students gain experience in experimental mechanics (test design, data acquisition), additive manufacturing (CAD, multi-material 3D printing with wrapping strategies), and data-driven optimization (Python, Bayesian optimization). *Communication* skills develop through lab notebooks, group presentations, conference talks, and co-authoring a peer-reviewed manuscript. *Professional* development includes project management, data-driven decision-making (interpreting optimization recommendations and selecting next experiments), interdisciplinary collaboration, and preparation for graduate school or industry careers.

6 Expected Research Outcomes

The primary outcome of this MRG is the **training of 2 undergraduate researchers** with hands-on experience in multi-material 3D printing, mechanical testing, and data-driven optimization, prepared for graduate study or engineering careers. By completion, each undergraduate is expected to have participated in the design and testing of approximately 100+ tensegrity specimens, contributed to at least one **conference presentation** at UCUR and ASME IDETC and contributed

as a co-author on a peer-reviewed manuscript submitted to the *ASME Journal of Mechanical Design* or *Smart Materials and Structures*. By centering the project on experimentation, students will each complete dozens of design–print–test cycles over two years, building deep practical competence through iterative, mentor-guided problem solving. The experimental data set—spanning a wide range of tensegrity-inspired geometries with measured energy-absorption performance—will be published on GitHub alongside optimization scripts for reproducibility. In Year 2, students will mentor incoming undergraduates, developing leadership skills and creating a self-sustaining research group.

7 Potential Impact of Work

This project advances architected material design for energy absorption with applications in **protective equipment** (helmets, body armor), **packaging** (shock-absorbing inserts), and **aerospace** (lightweight impact-resistant structures). By generating a large, systematically collected experimental data set of tensegrity-inspired structures, we provide the community with a resource for benchmarking and further optimization. The experiment-driven Bayesian optimization framework demonstrates that undergraduates can meaningfully drive a closed-loop experimental campaign, establishing a replicable mentoring model for data-driven research in other metamaterial classes. The layered mentoring structure—faculty, graduate co-mentor, and peer mentoring—can be adopted by other research groups seeking to involve undergraduates in hands-on engineering research. Open-source dissemination of data, code, and tested designs extends impact beyond BYU and enables other groups—including primarily undergraduate institutions—to build on this work.

8 Timeline

The project spans two academic years plus two summers (Table 1). **Year 1** focuses on experimental foundations: student recruitment and onboarding in Fall, followed by literature review, tensegrity parameterization in CAD, and initial multi-material print trials through Winter. The first Summer term is a critical period of full-time research with intensive mentoring—faculty increase meeting frequency to twice weekly, and the graduate co-mentor works alongside students daily. Students ramp up fabrication throughput, run systematic compression and drop-weight impact tests, and begin feeding results into the Bayesian optimization loop. **Year 2** shifts to optimization and dissemination: BO-guided experimental campaigns continue through Fall and Winter, with students printing and testing progressively refined designs. Students present findings at UCUR and ASME IDETC in Year 2, and a journal manuscript is prepared during Winter and Summer. Throughout, experienced students mentor newly recruited undergraduates, reinforcing their own skills and ensuring continuity of expertise.

9 Budget

Total requested: **\$25,000** over 2 years. Per grant guidelines, no funds are allocated to equipment, tuition, or faculty wages.

Task	Y1 F	Y1 W	Y1 Su	Y2 F	Y2 W	Y2 Su
Student recruitment & onboarding	•					
Literature review	•	•				
Tensegrity parameterization & CAD	•	•				
Print trials & process validation	•	•	•			
Multi-material 3D printing campaigns		•	•	•	•	
Compression & impact testing		•	•	•	•	
Bayesian optimization loop			•	•	•	
Conference presentations (UCUR/IDETC)				•	•	
Journal manuscript preparation					•	•
Peer mentoring of new students				•	•	

Table 1: Project timeline across two years (F = Fall, W = Winter, Su = Summer). Summer terms support full-time research with intensive mentoring. Year 2 includes formal peer mentoring, where experienced students supervise newly recruited undergraduates.

Category	Description	Year 1	Year 2
Undergraduate wages	2 students, part-time + summer full-time	\$9,000	\$9,000
Graduate student wages	Partial RA for grad co-mentor	\$1,250	\$1,250
Supplies	Filament, load cells, DAQ, consumables	\$2,250	\$750
Travel	Conference registration & travel	\$0	\$1,500
Annual Total		\$12,500	\$12,500

Undergraduate wages (\$18,000): Supports 2 undergraduates working part-time (~ 10 hrs/week) during fall and winter semesters and full-time (~ 40 hrs/week) during summer terms for simulation, 3D printing, and testing. **Graduate student wages (\$2,500):** Prof. Baird’s master’s student serves as day-to-day co-mentor. **Supplies (\$3,000):** PLA/TPU filament (sufficient for 50–100+ specimens), load cells, DAQ module, fasteners, poster printing, and other consumables (Year 1 heavy to support initial equipment and print trials). **Travel (\$1,500):** Student presentations at ASME IDETC or UCUR (Year 2).

References

- Ada Amendola, Everth Hernandez-Nava, Russell Goodall, Iain Todd, Hugh Shercliff, Hugh Shercliff, and Robert E. Skelton. Experimental investigation of the softening–hardening response of tensegrity prisms under compressive loading. *Composite Structures*, 117:234–243, 2014. doi: 10.1016/j.compstruct.2014.06.025.
- Sebastian Ament, Andrew Witte, Nishant Garg, and Julius Kusuma. Sustainable concrete via Bayesian optimization. *arXiv preprint arXiv:2310.18288*, 2023. URL <https://arxiv.org/abs/2310.18288>. NeurIPS 2023 Workshop on Adaptive Experimental Design and Active Learning in the Real World.
- Fernando Fraternali, Gerardo Carpentieri, and Ada Amendola. Tensegrity ring structures for impact protection. *International Journal of Solids and Structures*, 65–66:232–243, 2015. doi: 10.1016/j.ijsolstr.2015.03.024.

- Peter I. Frazier. A tutorial on Bayesian optimization. *arXiv preprint arXiv:1807.02811*, 2018. URL <https://arxiv.org/abs/1807.02811>.
- Nava Raj Khatri and Paul F. Egan. Energy absorption of 3D printed ABS and TPU multimaterial honeycomb structures. *3D Printing and Additive Manufacturing*, 11:e840–e850, 2024. doi: 10.1089/3dp.2022.0196.
- Changyu Mo, Paris Perdikaris, and Jordan R. Raney. Accelerated design of architected materials with multifidelity Bayesian optimization. *Journal of Engineering Mechanics*, 149(6):04023028, 2023. doi: 10.1061/JENMDT.EMENG-7033.
- Kirsti Pajunen, Paul Johanns, Raj Kumar Pal, Julian J. Rimoli, and Chiara Daraio. Design and impact response of 3D-printable tensegrity-inspired structures. *Materials & Design*, 182:107966, 2019. doi: 10.1016/j.matdes.2019.107966.
- Bobak Shahriari, Kevin Swersky, Ziyu Wang, Ryan P. Adams, and Nando de Freitas. Taking the human out of the loop: A review of Bayesian optimization. *Proceedings of the IEEE*, 104(1): 148–175, 2016. doi: 10.1109/JPROC.2015.2494218.
- David Silver, Aja Huang, Chris J. Maddison, Arthur Guez, Laurent Sifre, George van den Driessche, Julian Schrittwieser, Ioannis Antonoglou, Veda Panneershelvam, Marc Lanctot, et al. Mastering the game of Go with deep neural networks and tree search. *Nature*, 529(7587):484–489, 2016. doi: 10.1038/nature16961.
- Robert E. Skelton and Mauricio C. de Oliveira. *Tensegrity Systems*. Springer, New York, 2009. doi: 10.1007/978-0-387-74242-7.
- Felix Strieth-Kalthoff, Han Hao, Vandana Rathore, Joshua Derasp, Théophile Gaudin, Nicholas H. Angello, Martin Seifrid, Ekaterina Trushina, Mason Guy, Junliang Liu, et al. Delocalized, asynchronous, closed-loop discovery of organic laser emitters. *Science*, 384(6697):eadk9227, 2024. doi: 10.1126/science.adk9227.
- Cornel Sultan. Tensegrity: 60 years of art, science, and engineering. *Advances in Applied Mechanics*, 43:69–145, 2009. doi: 10.1016/S0065-2156(09)43002-3.
- Yue Wang et al. Bayesian optimization for the design of mechanical metamaterials. *Journal of the Mechanics and Physics of Solids*, 159:104734, 2022. doi: 10.1016/j.jmps.2021.104734.
- Haitao Ye, Qingjiang Liu, Jianxiang Cheng, Honggeng Li, Bingcong Jian, Rong Wang, Zechu Sun, Yang Lu, and Qi Ge. Multimaterial 3D printed self-locking thick-panel origami metamaterials. *Nature Communications*, 14:1607, 2023. doi: 10.1038/s41467-023-37343-w.
- Li-Yuan Zhang, Yue Li, Yan-Ping Cao, and Xi-Qiao Feng. Tensegrity cell mechanical metamaterial with metal rubber. *Applied Physics Letters*, 106(20):201901, 2015. doi: 10.1063/1.4921402.

Biographical Sketches

Jeffrey R. Hill — Principal Investigator

Professional Preparation

Ph.D.	Mechanical Engineering	University of Utah
M.S.	Mechanical Engineering	Brigham Young University
B.S.	Mechanical Engineering	Brigham Young University

Appointments

- Associate Professor, Department of Mechanical Engineering, Brigham Young University (current)
- Research Engineer, Sandia National Laboratories (prior)

Products Most Closely Related to This Proposal

Publications and products related to smart materials, shape memory alloys, compliant mechanisms, and structural mechanics.

Synergistic Activities

- Mentoring of undergraduate and graduate students in experimental mechanics and smart materials research at BYU.
- Teaching courses in mechanics of materials, machine design, and kinematics in the Department of Mechanical Engineering.
- Collaboration with national laboratories on advanced materials characterization and structural design.

Sterling G. Baird — Co-Principal Investigator

Professional Preparation

Ph.D.	Materials Science and Engineering	University of Utah
B.S.	Materials Science and Engineering	Brigham Young University

Appointments

- Faculty, Department of Mechanical Engineering, Brigham Young University (current)

Products Most Closely Related to This Proposal

Publications and products related to materials informatics, Bayesian optimization, data-driven materials discovery, and machine learning for materials science.

Synergistic Activities

- Development of open-source tools for Bayesian optimization and materials informatics.
- Mentoring of graduate and undergraduate students in data-driven methods for materials science and engineering.
- Active contributions to the materials informatics community through publications, software, and educational resources.